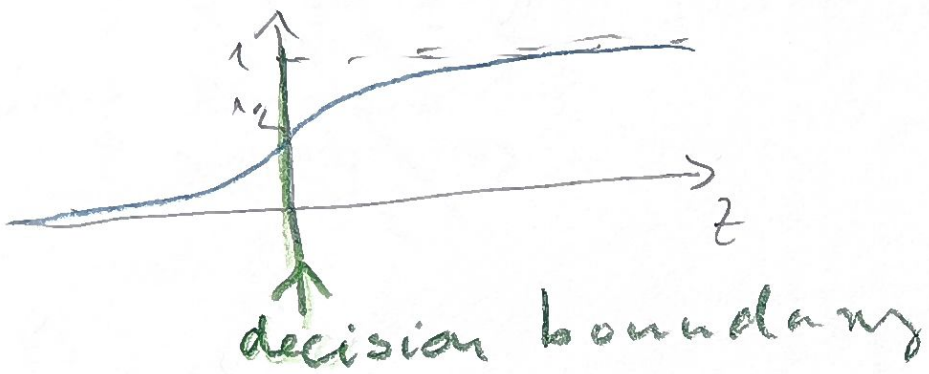
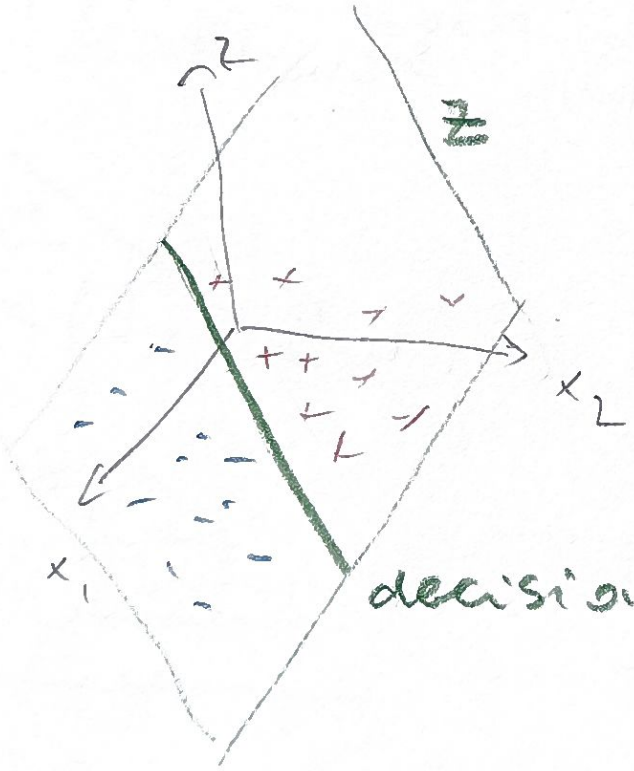
Introduction to machine learning -5-

Support Vector Machine (SVM)

Binary data, logistic regression, decision boundary

$z = W^T x + b$, $y = \sigma(z)$



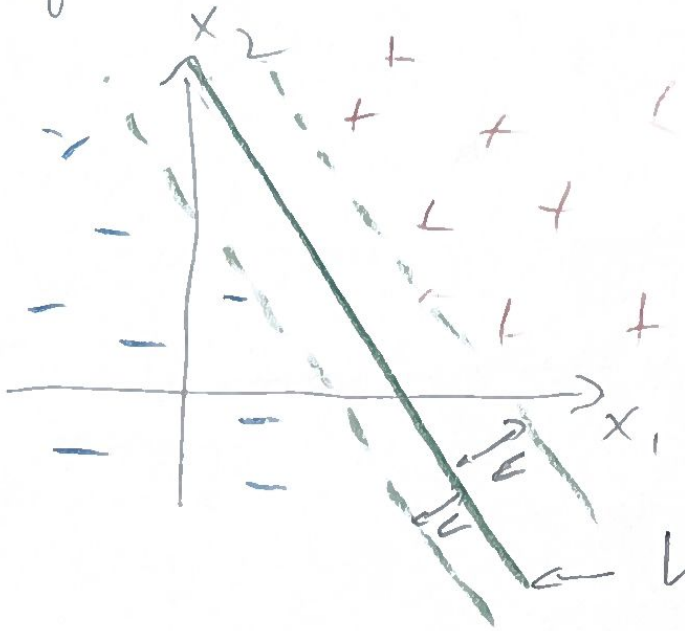


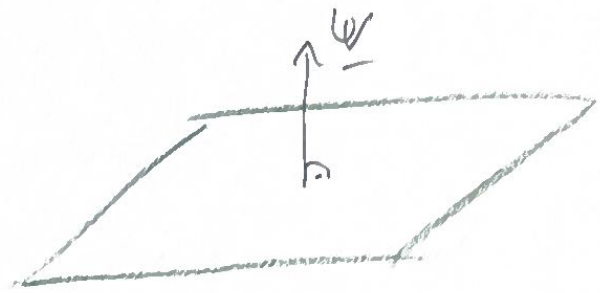
$W^T x + b = 0$



possible decision boundaries } which to choose?

Optimal separating hyperplane: maximize margin





hyperplane: $\underline{w}^T \underline{x} + b = 0$

↖ normal to the plane

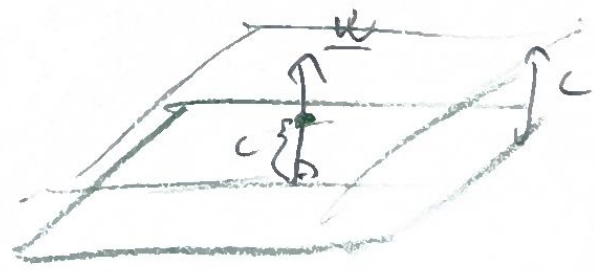
unit normal: $\underline{w}^* = \frac{\underline{w}}{|w|}$

distance from the plane:

$$D(x') = \frac{w^T x' + b}{|w|} = \underline{w}^* x' + \frac{b}{|w|} \quad \text{signed!}$$

Classification:

$$\text{sgn}(\underline{w}^T \underline{x}^{(i)} + b) = t^{(i)} \quad \text{or} \quad t^{(i)}(\underline{w}^T \underline{x}^i + b) > 0$$



Margin: c

$$t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) \cdot \frac{1}{|w|} \geq c$$

Optimization: choose $\underline{w}, b$ to

$$\max_{\underline{w}, b} c \quad \text{such that} \quad \frac{1}{|w|} t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) \geq c$$

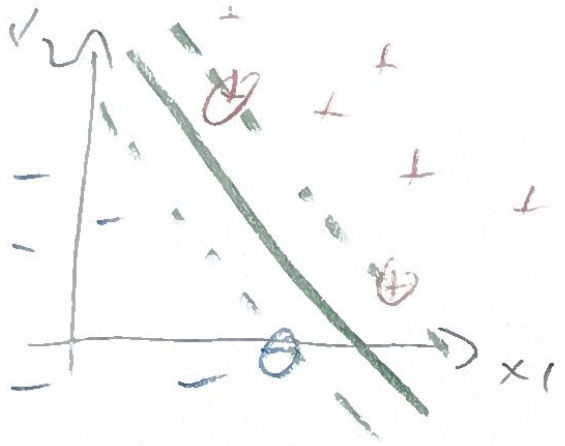
We have some freedom: $\underline{w}' = 2\underline{w}$ are equivalent

Choose: $\|W\| = \frac{1}{C} \Rightarrow t^{(i)}(\underline{W}^T \underline{x}^{(i)} + b) \geq 1$

↖ classification

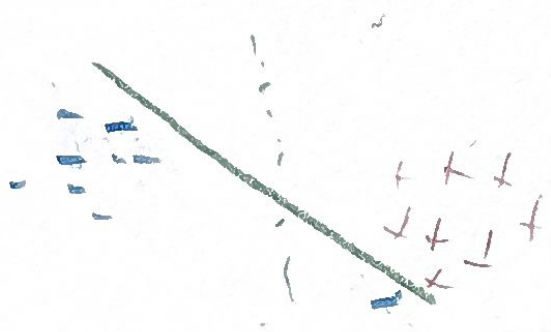
Optimization: minimize $\|\underline{W}\|^2$ for $\underline{W}, b$

such that $t^{(i)}(\underline{W}^T \underline{x}^{(i)} + b) \geq 1 \quad i = 1 \dots N$



only few data points determine $\underline{W}, b$. They are called the <u>support vectors</u>

<u>Problems:</u> noise, non-separable

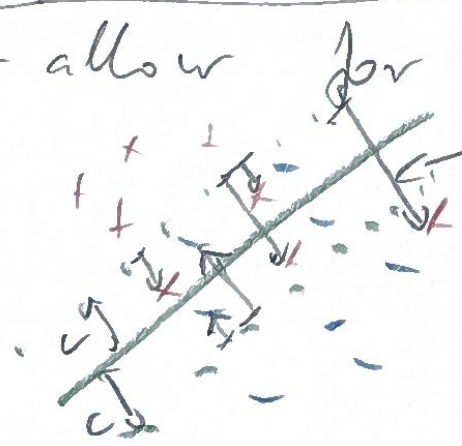




?

<u>Soft Margin SVM</u>

- allow for some error.



error $\xi^{(i)} > 0$

$0 < \xi^{(i)} < 1$: margin violation

$1 < \xi^{(i)}$: misclassification

Task: minimize $\xi^{(i)}$

↘

Soft margin SVM

$$\min_{\underline{w},b}\left(\|\underline{w}\|^2 + \lambda \sum_{i=1}^{N} \xi^{(i)}\right) \quad \text{such that}$$

$$t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) \geq 1 - \xi^{(i)}$$

λ hyperparameter

λ small: wide margin

λ large: large margin

Eliminate $\xi^{(i)}$. Two cases

$$1 - t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b)$$

≤ 0 ← → > 0

$\xi^{(i)} = 0$

$\xi^{(i)} = 1 - t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b)$

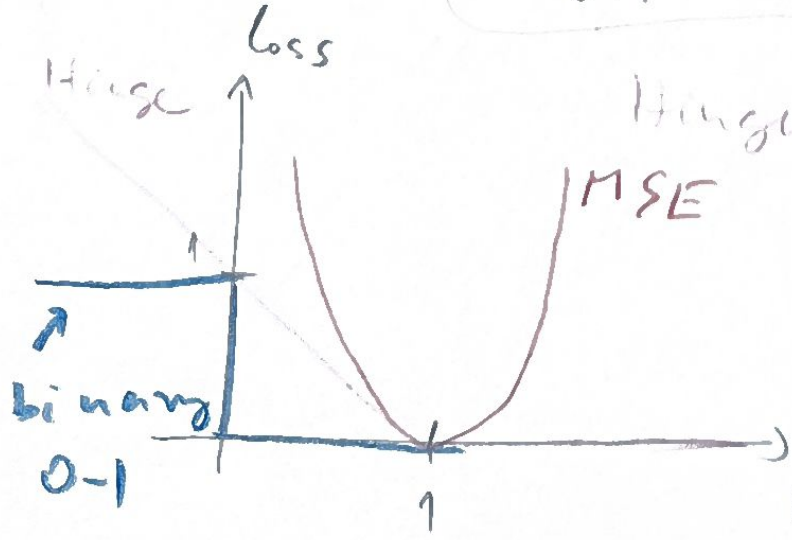
$$\xi^{(i)} = \max\left[0,\ t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b)\right]$$

Rewrite the function to be minimized which can be called loss function

$$\mathcal{L}_H(y,t) = \min_{\underline{w},b} \sum_{i=1}^{N} \max\left[0,\ 1 - t^{(i)} y^{(i)}(\underline{w},b)\right] + \underbrace{\frac{1}{2\lambda}\|\underline{w}\|^2}$$

Hinge loss

Regularization



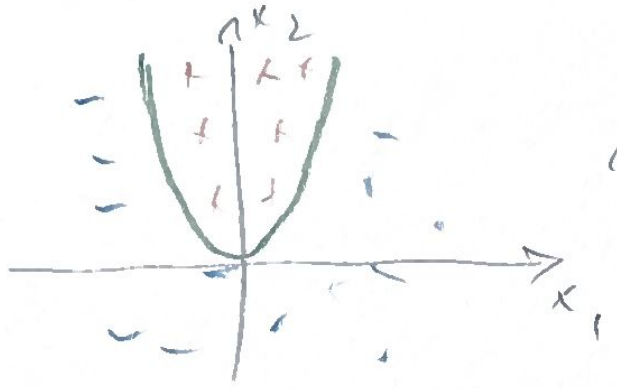
Soft margin SVM
linear classifier
with Hinge loss and
l_2 regularization!

IML - 5 -

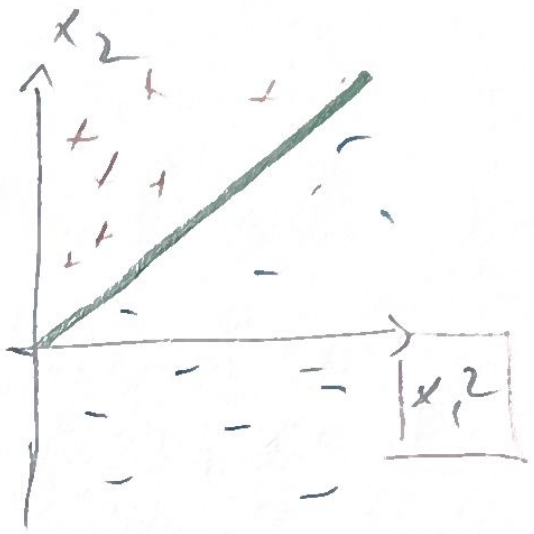
Kernel trick for non-linear boundary

Example:

data



$x_i' = \phi(x_i) = x_i^2$



$\underline{x}_i \longrightarrow \phi(\underline{x}_i)$

SVM: dual problem

$$\max_{\alpha_i \geq 0} \left[\sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{N} t^{(i)} t^{(j)} \alpha_i \alpha_j \left(\underline{x}^{(i)T} \underline{x}^{(j)} \right) \right]$$

$$\alpha_i \geq 0 \qquad \sum_{i=1}^{N} \alpha_i t^{(i)} = 0$$

Weights: $\underline{w} = \sum_{i=1}^{N} \alpha_i t^{(i)} x^{(i)}$

Much simpler quadratic problem

$\underline{x} \in \mathbb{R}^m$ Kernel $\phi(\underline{x}) \in \mathbb{R}^M$ $M >> m$

$\underline{x}^{(i)T} \underline{x}^{(j)} \rightarrow \phi(\underline{x}^{(i)})^T \phi(\underline{x}^{(j)}) =: K(\underline{x}^{(i)}, \underline{x}^{(j)})$

$\mathbb{R}^2$

Example: $\underline{x} = (x_1, x_2)$ $\quad \underline{z} = (z_1, z_2)$

$$\phi(\underline{x}) = \begin{pmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2}\, x_1 x_2 \\ \sqrt{2} x_1 \\ \sqrt{2} x_2 \\ 1 \end{pmatrix} \qquad \phi(\underline{z}) = \begin{pmatrix} z_1^2 \\ z_2^2 \\ \sqrt{2}\, z_1 z_2 \\ \sqrt{2} z_1 \\ \sqrt{2} z_2 \\ 1 \end{pmatrix}$$

$\uparrow$
$\mathbb{R}^6$

$$\phi(x)^T \phi(z) = x_1^2 z_1^2 + x_2^2 z_2^2 + 2 x_1 x_2 z_1 z_2 + 2 x_1 z_1 + 2 x_2 z_2 + 1 =$$

$$= (x_1 z_1 + x_2 z_2 + 1)^2$$

$$k(x, z) = \phi(x)^T \phi(z) = (x_1 z_1 + x_2 z_2 + 1)^2$$

Generally, people give $k(x, z)$ but it must obey rules.

Summary

+ Works well for high dimensional data
+ Best if more features than data
+ Robust to overfitting
+ Beautiful Kernel tricks for non-linear, non-separable data
- Not good for large datasets
- Deep learning is better in finding kernels